

High Levels of Visibility of Virtual Agents Increase the Social Presence of Users

Lucie Kruse*
Universität Hamburg

Fariba Mostajeran†
Universität Hamburg

Frank Steinicke‡
Universität Hamburg



Figure 1: The agent representations, from left to right: Mouth-Only, Head-Only, Upper Body, Full Body.

ABSTRACT

Virtual humanoid agents can perform a variety of tasks, and their representation has an influence on several psychological aspects. We compared the effects of agent visibility during an anagram solving task in virtual reality on social presence and cognitive task performance. We increased the visibility from (i) voice-only, (ii) mouth-only, (iii) full head, (iv) upper body, to (v) full body representations of the same virtual agent in a within-subject design study. The results revealed significant differences in the perceived social presence, especially for the two least visible representations, i.e., voice- and mouth-only, but no significant effects on task performance could be found.

Keywords: virtual reality, virtual agents, anthropomorphism, exergames, cognitive training

1 INTRODUCTION

Virtual agents (VA) have been integrated into various applications to play a variety of roles such as virtual coaches, assistants, and health professionals. VAs come in different shapes and representations, ranging from text and voice-interfaces only to fully-animated realistic 3D reconstructions of real humans. Previous research has shown that exposure to and interaction with VAs could result in *realistic*, i.e., similar-to-reality, social responses in humans. For instance, VAs can be used to elicit emotions such as social anxiety [5], irritation, and self-awareness [7] that humans usually demonstrate in response to other real humans.

Several studies have shown that the mere presence of a virtual agent could influence individuals' performance. This can be observed in the form of an improvement in task performance, which is known as *social facilitation* or a deterioration of the performance known as *social inhibition* [10]. However, the research on the effects of visual realism of virtual agents, in particular their level of visibility, on users' social presence or task performance is scarce. Therefore, we designed and conducted an experiment to study par-

ticipants' perception of social presence towards a virtual agent with varied levels of visibility and their task performance.

In particular, we hypothesized that higher levels of visibility lead to **(H1)** higher perceived social presence of the virtual agent. In addition, we expected to observe **(H2)** an increase in users' performance, e.g. more correct answers and a shorter response time, in easy tasks with an increase in the level of visibility (i.e., social facilitation effects). Similarly, we hypothesized that higher levels of virtual agent visibility lead to **(H3)** a decrease in users' performance in difficult tasks (i.e., social inhibition effect).

2 METHODS

In order to analyze these hypotheses, we implemented a virtual female agent in five different levels of visibility ranging from the lowest level (i) voice-only, (ii) mouth-only, (iii) full head, (iv) upper body to the highest level (v) full body (See Fig 1). In a within-subject study, participants solved anagrams in the presence of each representation of the agent in a virtual reality (VR) setting, which was implemented in Unity3d 2020.3.34. In order to study the effects on cognitive performance, we employed anagrams in the local language in three difficulty levels [3, 8]: (i) easy, (ii) medium and (iii) hard. Participants had to solve four easy, medium and hard words per condition. For each word, they had 40 seconds. We recorded several measurements, e.g. the number of correct words, and the time needed to solve each word.

During the study, the participants first answered demographic questions, and then put on the Pico 3 Pro Eye. They experienced each condition in a counterbalanced order. In each condition, the VA (in its corresponding representation) explained the task by saying: "Welcome to this round. Your task will [again] be to solve the following anagrams".

After each condition, we employed the Social Presence Survey (SPS) [1], and the *Social presence - Actor within medium (parasocial interaction)* subscale of the Temple Presence Inventory (TPI) [4]. After finishing all conditions, they answered additional questions about their favourite and least favourite VA representation. In total, the study took around one hour, with 20-25 minutes in VR.

3 RESULTS

Thirty participants between 19 and 34 years (26.33 ± 3.96 years) took part in the study (16 women, 13 men, one preferred not to say).

First, Shapiro tests were conducted to test for normal distribution for all data. Despite the partially non-normal distribution of the

*e-mail: lucie.kruse@uni-hamburg.de

†e-mail: fariba.mostajeran.gourtani@uni-hamburg.de

‡e-mail: frank.steinicke@uni-hamburg.de

data, we decided to report the analysis based on parametric tests (significance level = 0.05) in order to not switch between statistical tests [2]. The p-value of the pairwise comparisons were corrected using the Bonferroni method. Results can be seen in Table 1.

CONDITION	SPS	PARASOCIAL INTERACTION	FAVOURITE	LEAST FAVOURITE	TRAINING FAVOURITE
Full Body	0.19 ± 0.2	3.22 ± 0.22	8 (+1)	4	4 (+8)
Head	-0.18 ± 0.17	2.73 ± 0.18	2	3 (+1)	1 (+3)
Mouth-only	-0.92 ± 0.14	2.03 ± 0.15	1 (+1)	15 (+2)	0 (+3)
Upper Body	0.03 ± 0.12	3.28 ± 0.19	7(+1)	2 (+2)	4 (+4)
Voice-only	-1.68 ± 0.17	1.9 ± 0.14	10 (+1)	4	8 (+8)
Mixed	-	-	2	2	13

Table 1: Mean and SE from SPS and the Parasocial Interaction subscale of TPI, agent preferences regarding the favourite and least favourite representation, and long-term training favourite. Mixed results (participants chose two or more options) written in brackets.

3.1 Social Presence

A significant main effect of the virtual agent's representations on SPS was found ($F(4, 116) = 35.9917, p < 0.001, \eta_g^2 = 0.3935$). Pairwise comparisons reveal that voice-only and mouth-only conditions had significantly lower SPS values than all other conditions (voice-head, upperbody and full body: $p < 0.001$; voice-mouth: $p = 0.0055$; mouth-head: $p = 0.0013$; mouth-upper body and full body: $p < 0.001$); with voice-only receiving the lowest rating overall. No significant differences could be observed between full head and full body ($p = 0.2176$), and the upper body and full body or head ($p = 1$) conditions. In addition, we observed a significant main effect of the virtual agent's representations on Parasocial Interaction ($F(4, 116) = 31.6124, p < 0.001, \eta_g^2 = 0.2662$). Here, the difference between voice-only and mouth-only was not significant ($p = 1$), but all other conditions received significantly higher scores compared to voice-only and mouth-only conditions (voice-head: $p = 0.001$; voice-upper body or full body: $p < 0.001$; mouth-head and upper body and full body: $p < 0.001$). Furthermore, the upper body condition received significantly higher values than the full head ($p = 0.001$) condition. The difference between full head and full body ($p = 0.0511$) as well as upper body and full body ($P = 1$) were not significant. Thus, the results of social presence measures (i.e., Social Presence Survey and Parasocial Interaction) support H1.

3.2 Cognitive Performance

A 3 (task difficulty) x 5 (virtual agent's representations) repeated measures ANOVA was performed. No significant main effect of the agent's representation could be observed on the number of correct words ($F(4, 116) = 0.6652, p = 0.6174, \eta_g^2 = 0.0046$) or the time needed to solve all words ($F(4, 116) = 0.98, p = 0.4214, \eta_g^2 = 0.0062$). Thus, H2 and H3 cannot be confirmed.

3.3 User Preferences

Many people preferred the voice-only condition, for example since it provided the least distraction from the real task (P₉, P₁₁, P₁₇, P₂₅, P₂₈, P₃₀), whereas the two embodied conditions provided a feeling of a real person being there (P₅, P₁₀, P₁₉, P₂₀). The least favourite agent was the mouth-only representation, because it was "creepy" (P₃, P₉, P₂₄, P₂₇) and unnatural (P₈, P₂₀, P₂₅, P₂₈). Regarding the question about their preference for future cognitive training, many participants stated that it depended on the kind of task and duration (P₄, P₅, P₈, P₉, P₁₂, P₁₅, P₁₉, P₂₈).

4 DISCUSSION AND CONCLUSION

Our results revealed that the perceived social presence increased with higher level of agent visibility, which is in line with prior re-

search [9]. An especially large difference could be observed between the two conditions without eyes, namely voice-only and mouth-only representations. They showed significantly lower social presence than the other conditions, indicating that the interaction with them did not feel as human-like as with the other conditions. In addition, only the upper body and full body representations were in a positive range of the social presence survey, implying that they seemed the most conscious and human-like to the participants compared to the rest of representations. However, both social presence measures did not present very high ratings for any of the agent representations. This might be because of the limited interaction the users had with the agent representations. During each condition, the agent only explained the task at the start and did not speak to the user again afterwards.

We could not find any significant differences between task performance in the presence of the five representations of the VA. Thus, in contrast to our priori hypotheses, we could not observe any social facilitation and inhibition effects on the anagram solving task. This might be due to several reasons. First, the difficulty of the task we used might not have been suitable to show these effects. For the anagrams we chose, most of the easy ones could be solved by all participants, and most of the hard anagrams were too hard to be solved at all. This might have led to floor/ceiling effects. Furthermore, the chosen cognitive task might not reveal any social facilitation or inhibition effects. However, a verbal task in which the VA played a more visible role could cause a difference. For example, if the correct word would have been verbalized to the VA, as done by Mostajeran et al. [6], the VA would have been more in focus.

As a summary, high levels of agent visibility have a significant positive effect on the perceived social presence. Regarding the cognitive tasks, the results did not show any effects on task performance. The results provide important implications for the design of future VAs for innovative technologies, such as AR/VR systems or assistants in smart homes or new working or learning environments.

REFERENCES

- [1] J. Bailenson and J. Blascovich. Interpersonal distance in immersive virtual environments. *Personality & social psychology bulletin*, 29:819–33, 08 2003. doi: 10.1177/0146167203029007002
- [2] M. J. Blanca Mena, R. Alarcón Postigo, J. Arnau Gras, R. Bono Cabré, R. Bendayan, et al. Non-normal data: Is anova still a valid option? *Psicothema*, 2017.
- [3] A. Hoffmann, B. Diedenhofen, B. Verschuere, and J. Musch. A strong validation of the crosswise model using experimentally-induced cheating behavior. *Experimental Psychology*, 2015.
- [4] M. Lombard, T. B. Ditton, and L. Weinstein. Measuring presence: the temple presence inventory. In *Proceedings of the 12th annual international workshop on presence*, pp. 1–15, 2009.
- [5] F. Mostajeran, M. B. Balci, F. Steinicke, S. Kühn, and J. Gallinat. The effects of virtual audience size on social anxiety during public speaking. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 303–312. IEEE, 2020.
- [6] F. Mostajeran, N. Burke, N. Ertugrul, K. Hildebrandt, J. Matov, N. Tapie, W. G. Zittel, P. Reisewitz, and F. Steinicke. Anthropomorphism of virtual agents and human cognitive performance in augmented reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 329–332. IEEE, 2022.
- [7] M. Slater and M. Usoh. Body centred interaction in immersive virtual environments. *Artificial life and virtual reality*, 1(1994):125–148, 1994.
- [8] A. Voss. *Motivierte Wahrnehmung: selektive Aufmerksamkeit und entlastende Umdeutungen bei der Aufnahme valenter Informationen*. PhD thesis, Trier, Univ., Diss., 2004, 2004.
- [9] B. Yoon, H.-i. Kim, G. A. Lee, M. Billinghurst, and W. Woo. The effect of avatar appearance on social presence in an augmented reality remote collaboration. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 547–556. IEEE, 2019.
- [10] R. B. Zajonc. Social facilitation. *Science*, 149(3681):269–274, 1965.